

KHRABROVO, Russian Federation

WMO: 267020

Lat: **54.890N**Lon: **20.593E**Elev: **43**StdP: **14.67**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.5	8.5	-0.8	5.3	5.1	4.3	6.9	10.6	30.8	37.4	26.2	38.0	4.3	110	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	83.0	68.3	79.3	66.4	76.4	65.0	70.6	79.1	68.5	76.5	66.6	74.0	7.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	102.4	75.0	65.8	95.3	72.5	64.0	89.3	70.5	34.5	78.9	32.7	76.5	31.2	73.8	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.5	16.5	DB	-3.5	89.2	8.0	3.5	-9.2	91.7	-13.9	93.7	-18.3	95.7	-24.1	98.2	(4)
(5)			WB	-3.6	74.8	7.6	2.4	-9.1	76.5	-13.5	77.9	-17.8	79.3	-23.3	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	28.7	30.3	35.9	45.7	54.3	60.3	65.1	64.3	56.4	47.3	39.1	32.4	(6)	
	DBStd	14.67	10.02	9.62	7.13	7.25	6.94	5.38	5.15	4.74	5.26	6.39	7.07	9.17	(7)	
	HDD50	2862	661	551	438	169	31	0	0	0	10	128	327	547	(8)	
	HDD65	6811	1126	971	902	580	339	162	63	72	260	548	776	1012	(9)	
	CDD50	1677	0	0	1	40	166	309	469	443	203	44	2	0	(10)	
	CDD65	147	0	0	0	0	9	20	66	49	3	0	0	0	(11)	
	CDH74	1230	0	0	0	19	108	205	506	362	30	0	0	0	(12)	
	CDH80	276	0	0	0	1	13	47	142	70	3	0	0	0	(13)	
(14) Wind	WSAvg	7.2	8.3	7.9	7.8	7.1	6.7	6.6	6.2	5.9	6.2	7.2	7.9	8.3	(14)	
(15) Precipitation	PrecAvg	30.8	2.4	1.9	1.9	1.5	2.0	2.7	3.5	3.5	2.8	3.0	3.0	2.6	(15)	
	PrecMax	43.1	5.7	3.1	4.0	3.1	4.3	5.1	7.8	7.6	5.8	6.8	6.1	3.9	(16)	
	PrecMin	22.6	0.4	0.5	0.2	0.0	0.6	0.4	0.4	0.2	0.6	0.6	1.0	1.5	(17)	
	PrecStd	4.3	1.1	0.7	1.0	0.7	0.9	1.1	1.9	1.8	1.4	1.6	1.4	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	49.1	59.5	76.7	82.0	85.9	89.3	86.3	77.9	66.6	55.1	50.2	(19)	
		MCWB	45.3	43.7	47.9	60.7	65.4	70.3	70.9	70.5	64.8	58.7	51.5	47.7	(20)	
	2%	DB	44.6	46.2	53.4	69.4	77.0	80.1	84.1	81.0	72.4	62.5	52.0	47.2	(21)	
		MCWB	42.5	42.8	45.7	56.0	62.9	66.7	69.3	67.2	62.7	56.5	49.1	45.3	(22)	
	5%	DB	42.4	43.1	48.6	63.1	72.0	75.3	79.3	77.8	68.9	59.2	50.1	45.0	(23)	
		MCWB	40.5	40.8	43.5	51.8	60.0	63.7	67.5	65.8	60.6	55.0	47.8	43.4	(24)	
	10%	DB	39.9	41.1	45.6	58.6	67.6	71.5	75.4	74.9	66.1	56.9	48.2	43.0	(25)	
		MCWB	38.4	39.2	41.6	49.7	57.6	61.8	65.8	64.7	59.2	53.6	46.2	41.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.8	45.4	50.4	61.3	67.8	71.8	74.5	72.9	67.1	60.2	52.3	48.5	(27)	
		MCDB	47.0	47.1	55.9	75.3	79.0	81.8	84.2	81.6	74.6	64.5	54.4	49.7	(28)	
	2%	WB	42.6	43.0	46.8	56.5	64.3	68.0	71.2	69.8	64.1	57.5	49.8	45.7	(29)	
		MCDB	44.0	45.1	51.3	67.4	74.4	77.7	79.9	78.1	69.8	61.1	51.4	46.8	(30)	
	5%	WB	40.3	41.1	44.1	53.2	61.5	65.3	69.0	67.7	62.0	55.6	48.1	43.6	(31)	
		MCDB	41.8	43.1	47.9	61.3	70.1	73.2	76.6	74.7	67.3	58.6	49.7	44.8	(32)	
	10%	WB	38.4	39.2	41.7	50.4	58.7	62.9	67.0	65.8	60.1	53.9	46.2	41.4	(33)	
		MCDB	39.7	40.8	44.8	56.8	66.0	69.4	74.3	72.1	64.7	56.7	47.8	42.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.6	12.8	18.0	19.0	17.5	16.8	17.3	15.9	12.5	8.3	7.9	(35)	
		MCDBR	8.1	9.7	18.6	27.2	26.0	25.2	23.5	23.1	21.4	16.6	10.2	7.8	(36)	
	5% WB	MCWBR	7.0	7.6	12.4	15.4	13.6	13.4	11.3	11.6	12.6	11.2	8.4	7.2	(37)	
		MCWBR	7.8	9.1	17.1	25.1	24.0	22.6	21.2	20.0	19.1	14.6	9.3	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.311	0.338	0.367	0.408	0.395	0.395	0.426	0.419	0.378	0.359	0.346	0.309	(40)	
		taud	2.254	2.265	2.262	2.218	2.326	2.362	2.292	2.311	2.400	2.410	2.299	2.253	(41)	
	Ebn at Noon	Ebn at Noon	194	232	252	258	269	271	258	252	248	224	178	166	(42)	
		Edh at Noon	19	26	33	39	37	36	38	36	29	23	18	15	(43)	
(44) All-Sky Solar Radiation	RadAvg		160	384	789	1319	1733	1795	1648	1379	977	488	181	110	(44)	
	RadStd		25	64	98	166	156	134	178	140	111	77	21	12	(45)	

Historical Trends

		DBAvg
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